

Statistics: how far the prices can be trusted

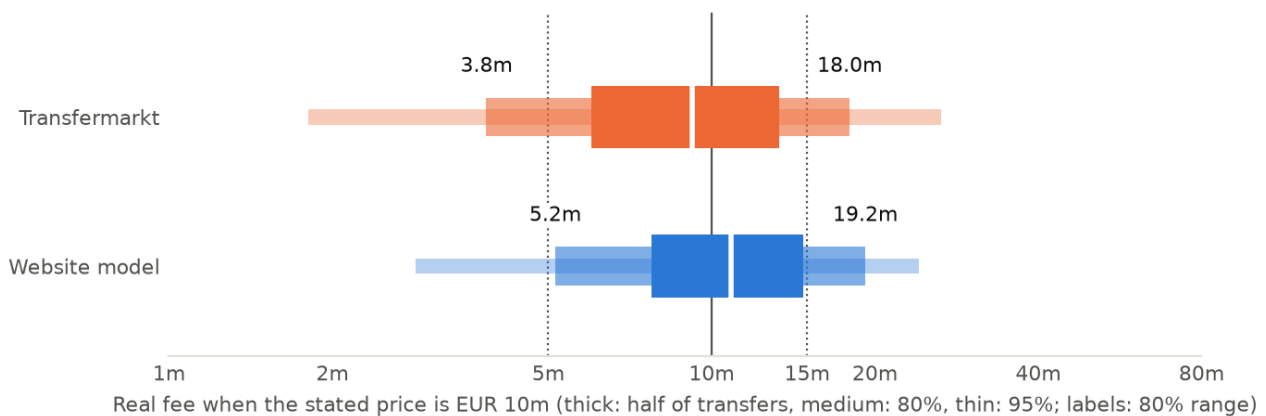
Working document, updated at each step. Last update: 4 October 2026.

1. The question and the short answer

A price source says a player is worth EUR 10m. If he is actually sold, how far from 10m can the fee be?

The answer below is measured on 5,252 real transfers from 2024 to mid-2026 that the model never saw in training. The rows use the 1,043 transfers that Transfermarkt priced between EUR 5m and 20m and the 823 that the website model priced in that band, because the spread depends on the price level (section 6).

Stated price: EUR 10m	Transfermarkt	Website model
Half of real fees fall between	6.0m to 13.3m	7.7m to 14.8m
80% of real fees fall between	3.8m to 18.0m	5.2m to 19.2m
95% of real fees fall between	1.8m to 26.5m	2.8m to 24.1m
Real fee inside 5m to 15m (within 50%)	65%	66%
Real fee below 5m	18%	9%
Real fee above 15m	17%	24%
Typical real fee (median)	9.2m	10.9m
Top of the 80% range divided by its bottom	x4.7	x3.7
Standard deviation (log scale)	0.68	0.57



So a range of 5m to 15m around a 10m price holds for 65% of players with Transfermarkt and 66% with the website model. Across all price levels the figures are 52% and 56%: the spread is wider for cheap players.

In this band the model's range is narrower in proportion (standard deviation 0.57 against 0.68) but it sits higher: the typical fee is 10.9m for a 10m model price, so the model misses mostly on the high side (24% of fees above 15m, 9% below 5m), where Transfermarkt misses on both sides.

2. How the error is measured

- **Error.** For each transfer, $e = \ln(\text{real fee} / \text{stated price})$. $e = 0$ means the price was exact, $e = 0.69$ means the fee was twice the price, $e = -0.69$ means half. Logs are used because price errors are proportional: being 5m off matters at 10m and not at 100m.
- **Mean.** The average of e . It is the systematic bias: above 0, the source prices too low.

- **Variance and standard deviation.** $s^2 = \text{sum of } (e - \text{mean})^2 / (n - 1)$, and s is its square root. s is the typical distance between a fee and the price. $\exp(s)$ turns it into a factor: about two thirds of fees lie between $\text{price} / \exp(s)$ and $\text{price} \times \exp(s)$ if errors are bell-shaped.
- **Standard error.** $s / \text{square root of } n$. It is the uncertainty on the mean, not on one player. With 5,252 transfers it is small, which is why averages are reliable and single prices are not.
- **Ranges.** The 50%, 80% and 95% ranges are read directly from the data (percentiles of fee / price), with no assumption on the shape of the errors.
- **Transfermarkt price.** The last value published at least 30 days before the transfer, so that it cannot already contain the fee.
- **Model price.** The website model (no buyer information), trained on the 9,886 transfers before 2024 and applied to 2024-2026. The full model, which knows the buyer, is shown for comparison.

3. Transfermarkt and the model: full statistics

All 5,252 test transfers. Log scale: statistics of $e = \ln(\text{fee} / \text{price})$.

Log scale	Transfermarkt	Website model	Full model
Mean (bias)	-0.133	+0.007	-0.010
Standard error of the mean	0.013	0.010	0.009
95% confidence interval of the mean	-0.158 to -0.108	-0.013 to +0.028	-0.029 to +0.008
Variance	0.870	0.575	0.452
Standard deviation	0.933	0.758	0.673
95% confidence interval of the standard deviation	0.909 to 0.958	0.740 to 0.777	0.655 to 0.692
Standard deviation as a factor, $\exp(s)$	x2.54	x2.13	x1.96
Root mean squared error	0.942	0.758	0.673
Skewness	+0.25	-0.26	-0.48
Excess kurtosis	1.84	1.52	1.78

Same transfers, in plain ratios: real fee divided by stated price.

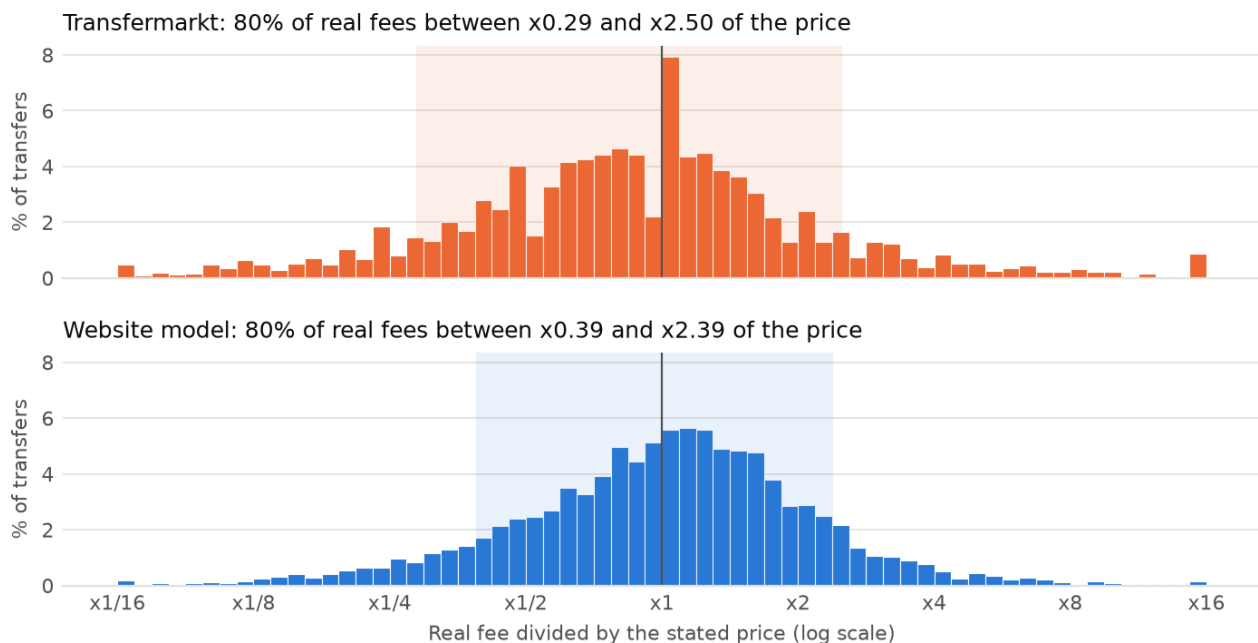
Ratio fee / price	Transfermarkt	Website model	Full model
Median	0.88	1.06	1.04
Mean	1.50	1.35	1.23
Variance	12.72	2.16	0.93
Standard deviation	3.57	1.47	0.97
Standard error of the mean	0.049	0.020	0.013
Middle 50% of transfers	0.50 to 1.43	0.65 to 1.60	0.68 to 1.52
Middle 68% (one standard deviation)	0.39 to 1.88	0.50 to 1.97	0.55 to 1.81
Middle 80%	0.29 to 2.50	0.39 to 2.39	0.43 to 2.10
Middle 95%	0.13 to 6.56	0.20 to 4.18	0.22 to 3.26

Ratio fee / price	Transfermarkt	Website model	Full model
Fee within 25% of the price	28%	30%	34%
Fee within 50% of the price	52%	56%	61%
Median absolute error, in % of the price	47%	44%	39%

The method note reports median errors of 50% and 42%. Those are measured in % of the fee; here the error is in % of the stated price, which is what the 10m example needs.

4. Reading the numbers

- **Transfermarkt.** Standard deviation 0.93 in logs, a factor of 2.54. The mean is -0.133 with a standard error of 0.013: on average fees are below Transfermarkt values (-12% at the median), and this bias is statistically clear.
- **Website model.** Standard deviation 0.76, a factor of 2.13. The variance falls from 0.87 to 0.58, so the model removes 34% of the variance of Transfermarkt's error. Its mean is +0.007 with a standard error of 0.010: no measurable bias.
- **The plain standard deviation misleads.** In ratios the standard deviation is 3.57 for Transfermarkt and 1.47 for the model, which would put a 10m player at 10m plus or minus 35.7m and 14.7m. A few transfers at several times the price inflate these figures; the percentile ranges are the ones to use.
- **Errors are not bell-shaped.** Excess kurtosis is 1.8 for Transfermarkt and 1.5 for the model (0 for a normal curve): extreme misses are more frequent than a normal curve predicts. This is why the ranges are taken from percentiles.



5. Is the model really better than Transfermarkt?

Both are compared on the same 5,252 transfers, so the test is paired: for each transfer, the absolute log error of Transfermarkt minus that of the model.

Paired comparison with Transfermarkt	Website model	Full model
Mean reduction in absolute log error	0.120	0.186

Paired comparison with Transfermarkt	Website model	Full model
Standard error	0.007	0.007
95% confidence interval	0.107 to 0.133	0.172 to 0.200
Ratio to standard error (t)	17.7	25.1
Transfers where the model is closer	58%	62%

The website model is closer to the real fee than Transfermarkt on 58% of transfers. The gain is far beyond chance, but on a single player it is not guaranteed: Transfermarkt is closer on the other 42%.

6. The spread depends on the price level

Transfers are grouped by the price each source gave. Standard deviation in logs, then the 80% range of fee / price.

Stated price (EUR)	n	Std dev	80% range	Within 50%	n	Std dev	80% range	Within 50%
under 1m	2216	1.10	0.27 to 4.00	43%	2293	0.85	0.33 to 2.74	50%
1m to 5m	1767	0.85	0.25 to 2.00	51%	1903	0.74	0.41 to 2.43	53%
5m to 20m	1043	0.68	0.38 to 1.80	65%	823	0.57	0.52 to 1.92	66%
over 20m	226	0.52	0.59 to 1.44	87%	233	0.45	0.63 to 1.43	90%

Columns 2 to 5: Transfermarkt. Columns 6 to 9: website model.

7. Skill or luck?

The test of luck: keep the same 5,252 prices and the same 5,252 fees, but hand the prices to the players at random, 5,000 times. If the real prices do no better than the shuffled ones, they are luck.

Against luck	Transfermarkt	Website model	Full model
Share of fee differences explained (R2)	64%	77%	82%
Same, prices shuffled: average	-79%	-77%	-80%
Same, prices shuffled: best of 5,000	-68%	-69%	-71%
Distance from luck, in standard deviations	58	62	65
Probability of doing this well by luck	p < 0.001	p < 0.001	p < 0.001
Typical error (RMSE, log scale)	0.94	0.76	0.67
Typical error, prices shuffled	2.09	2.08	2.09
Typical error, same price for every player	1.57	1.57	1.57
Correlation of price and fee (logs)	0.81	0.88	0.90
Rank correlation	0.80	0.87	0.90

Neither source is luck. In 5,000 reshuffles, none came close to the real prices: a shuffled price explains less than nothing (it is worse than giving every player the same price), Transfermarkt explains 64% of the differences in fees and the website model 77%. What is left unexplained, 36% and 23%, is the part that is not yet under control.

8. Is the model's correction skill or luck?

Beating luck is easy for the model, because it starts from the Transfermarkt value. The harder question is whether the correction it applies to that value is information or noise. Three tests on the same 5,252 transfers.

The correction to the Transfermarkt value	Website model	Full model
Direction right (fee on the side of Transfermarkt the model said)	71%	76%
Probability of that by coin flip	$p < 0.001$	$p < 0.001$
Real correction per unit of predicted correction (1 = right size)	0.96 (std error 0.02)	1.00 (std error 0.02)
t-statistic against no information (0)	44.2, $p < 0.001$	59.8, $p < 0.001$
Correlation of predicted and real correction	0.58	0.69
Squared error lower than Transfermarkt: t-statistic	19.2, $p < 0.001$	23.9, $p < 0.001$

Luck does not repeat. The same comparison year by year, website model against Transfermarkt:

Year	Transfers	Error, Transfermarkt	Error, website model	t-statistic	Probability by luck
2024	2,037	0.94	0.75	11.4	$p < 0.001$
2025	2,355	0.93	0.75	11.4	$p < 0.001$
2026	860	0.98	0.78	7.3	$p < 0.001$

The correction points the right way on 71% of transfers against 50% for a coin, and the model beats Transfermarkt in each of the 3 years. It is skill. It is also far from complete: the correlation between predicted and real correction is 0.58, so most of the gap between fee and Transfermarkt value is still unexplained.

9. What statistically right means, and where each source stands

Significant is not the same as right. A price source is statistically right when it passes all of the tests below at the 5% level. Tests 2, 3, 4, 7 and 8 are passed when no flaw can be detected; tests 1, 5 and 6 when the advantage cannot be luck. This scorecard is the target for the next steps.

Criterion	Test	Transfermarkt	Website model	Full model
1. Skill, not luck	Prices explain fees better than the same prices shuffled	Pass $p < 0.001$	Pass $p < 0.001$	Pass $p < 0.001$
2. No overall bias	Mean log error is 0 (t-test)	Fail fees -12%, $p < 0.001$	Pass fees +1%, $p = 0.475$	Pass fees -1%, $p = 0.260$
3. Right scale	Fee rises one for one with price (slope 1, intercept 0)	Fail slope 0.91, $p < 0.001$	Pass slope 1.00, $p = 0.770$	Pass slope 1.01, $p = 0.244$
4. No bias at any price level	Mean log error is 0 in each of the four price bands (worst band shown)	Fail 1m to 5m: fees -24%, $p < 0.001$	Pass 1m to 5m: fees +3%, $p = 0.319$	Fail under 1m: fees -4%, $p = 0.020$
5. Beats Transfermarkt	Squared error lower on the same transfers	reference	Pass $t = 19.2$, $p < 0.001$	Pass $t = 23.9$, $p < 0.001$

Criterion	Test	Transfermarkt	Website model	Full model
6. Beats it every year	Same test in 2024, 2025, 2026	reference	Pass weakest year: t = 7.3	Pass weakest year: t = 10.4
7. 80% range is honest	Range set on 2023 holds 80% of 2024-2026 fees	no range published	Pass holds 80.5%, p = 0.388	Fail holds 81.2%, p = 0.031
8. 90% range is honest	Same at 90%	no range published	Pass holds 89.5%, p = 0.223	Pass holds 90.4%, p = 0.370

Score: Transfermarkt 1 of 4, website model 8 of 8, full model 6 of 8.

Bias by price band (mean error as fees above or below the price; p-values corrected for testing four bands at once):

Stated price (EUR)	Transfermarkt	Website model	Full model
under 1m	-1%, p = 0.653	-1%, p = 0.978	-4%, p = 0.020
1m to 5m	-24%, p < 0.001	+3%, p = 0.319	+2%, p = 0.671
5m to 20m	-15%, p < 0.001	+2%, p = 0.978	+2%, p = 0.671
over 20m	-9%, p = 0.020	-3%, p = 0.978	-3%, p = 0.671

- **Transfermarkt** has real skill but is not statistically right: its slope is 0.91 instead of 1 and fees differ from its values by -12% on average.
- **The website model** has skill beyond Transfermarkt, no detectable overall bias and a slope of 1.00 (not distinguishable from 1). Its 80% range holds 80.5% of fees and its 90% range 89.5%.
- **Mean and median disagree for the model.** The tests above are on the mean of the log error. The median fee is above the model price (54% of fees are above it, sign test p < 0.001): the model is right on average because a minority of large overpricings offsets a majority of small underpricings.
- **Being right on average is not being precise.** A model can pass every test here and still have wide ranges. The tests say whether the prices and ranges can be believed; the standard deviation of section 3 says how useful they are. Both must improve.

10. Expensive players: the correction went too far

For a player Transfermarkt values highly, the model stacked bonuses (young, sold from England, expensive last transfer) that the Transfermarkt value already contains. The check: when the model predicts a correction, how much of it shows up in the real fee? 1 means all of it. Below, the website model on the 5,252 unseen transfers, before and after the fix.

Transfermarkt value	Transfers	Real correction per unit predicted: before	after	Typical error: before	after	Transfermarkt
under 5m	3,983	0.98	0.98	0.803	0.803	1.013
5m to 20m	994	0.74	0.80	0.630	0.626	0.713
20m to 40m	205	0.57	0.78	0.516	0.498	0.538
over 40m	70	0.43	0.66	0.402	0.374	0.388

- **Before.** The correction was reliable for cheap players and too strong for expensive ones. On the 36 transfers valued at 20m or more where the model announced at least 1.3 times the Transfermarkt value, it announced a median of 1.55 times and the real fees came in at 1.30 times.
- **The fix.** The correction is scaled down as the Transfermarkt value rises: untouched below EUR 5m, multiplied by 0.65 above EUR 40m, in between in proportion. These three numbers are estimated on the training transfers only, with predictions made out of sample.
- **After.** On the same 36 transfers the model now announces 1.33 times. Over the whole test set the gain in squared error has a t-statistic of 4.5. Above 40m the model's error is 0.374 against 0.388 for the Transfermarkt value on its own: the model adds something even for the most expensive players.
- **Ranges.** The width of the 80% and 90% ranges is now set separately for four levels of Transfermarkt value (under 1m, 1m to 5m, 5m to 20m, over 20m), because errors are much smaller for expensive players. Tests 7 and 8 of the scorecard use these ranges.

11. Limits of these statistics

- **A fee is not the true value.** Two clubs can agree on different fees for the same player depending on urgency, contract length or the buyer. Part of the spread is this deal noise, which no price can remove. The standard deviations above are an upper bound on the pricing error itself.
- **Only sold players are measured.** The statistics describe players who were transferred for a disclosed fee of at least EUR 50,000. Players who are not for sale may behave differently.
- **The floor at EUR 50,000** cuts the lowest fees, which narrows the measured spread for players priced under 1m.
- **One test period.** 2024 to mid-2026. The spread may differ in other market conditions.
- **Transfers are treated as independent.** Fees in the same window move together, so the true p-values are somewhat larger than shown. This cannot change results with t above 10; it matters for the borderline ones.
- **With 5,252 transfers, small flaws become significant.** A failed test says a flaw exists, not that it is large. The size is given next to each test.

Update log

- **4 October 2026.** Step 1. Spread of real fees around the Transfermarkt value and around the model price: variance, standard deviation, standard error, ranges, worked example at EUR 10m.
- **4 October 2026.** Step 2. Significance tests: skill against luck, bias, calibration, stability by year, coverage of the ranges, and a scorecard defining what statistically right means.
- **4 October 2026.** Step 3. Model changed: eleven position statistics added (54 parameters instead of 43 for the website model). All figures in this document recomputed with the new model.
- **4 October 2026.** Step 4. Model changed: the correction is scaled down for expensive players and the ranges are set by level of Transfermarkt value (section 10). All figures recomputed.